INF201 Deliverable 3

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Class overview

Dice

An n-sided die used to introduce randomness

Attributes

- `sides: int` : Number of sides (default 6, must be >=2, otherwise `ValueError`)

Methods

- `roll()` : Returns a random integer from 1 to `sides``
- `__call__()` : Lets you write `d20()` instead of `d20.roll()`
- `__repr__()` : Returns e.g. `Dice(sides=6)`

Weapon

An item a character can equip. Damage comes from a `Dice` object

Attributes

- `name: str` : Name of the weapon
- `damage_dice: Dice` : The die rolled to determine damage

Methods

- `roll_damage()` : Rolls `damage_dice` and returns the damage as an int

Character

The base class for all fighters

Attributes

- `name: str` : Character's name
- `_hp: int` : Current HP (protected, accessed via the `hp` property)
- `hp_max: int` : Maximum HP (default 100)
- `damage: int` : Base damage without a weapon (default 10)
- `weapon: Weapon = None` : Optional equipped weapon

Methods

- `attack(target)` : Rolls weapon damage if equipped, otherwise uses `damage` , then calls `target.take_damage()``
- `take_damage(damage)` : Reduces own HP, never below 0
- `heal(heal)` : Increases own HP, never above `hp_max`
- `introduce()` : Prints a greeting
- `hp` (property): Read-only access to `_hp``
- `is_alive` (property): `True` if `_hp > 0`
- `__repr__()` & `__str__()` : Developer and readable string representations

Warrior(Character)

Subclass of Character. A defensive specialist.

Added attributes

- `armor: int` : Armor rating (default 5)

Overridden methods

- `take_damage(damage)` : Reduces incoming damage by `armor` (minimum 0) before applying it.

Mage(Character)

Subclass of Character. A resource-based attacker.

Added attributes

- `mana: int` : Current mana (default 30)
- `mana_max: int` : Maximum mana (default 30)

Overridden methods

- `attack(target)` : With >=10 mana: spends 10 mana and deals double damage, ignoring weapons. Otherwise: base attack with the equipped weapon and restores 5 mana

Battle

Manages turn-based combat between two characters.

Attributes

- `hero: Character` : The hero
- `enemy: Character` : The enemy

Methods

- `play_round()` : Hero attacks; if the enemy is still alive, the enemy attacks back
- `simulate_battle()` : Repeats rounds until one character is no longer alive

Battle simulations

In [8]:

```python
# Import classes
from battle import Battle
from character import Character
from dice import Dice
from items import Weapon
from mage import Mage
from warrior import Warrior

# Create weapons
dagger     = Weapon("Dagger", Dice(6))
staff      = Weapon("Staff", Dice(6))
sword      = Weapon("Sword", Dice(8))
axe        = Weapon("Axe", Dice(10))
warhammer  = Weapon("Warhammer", Dice(20))

# Create characters
alice = Character("Alice")
bob = Character("Bob", hp_max=80, damage=12)
goblin = Character("Goblin", hp_max=60, damage=8, weapon=dagger)
ogre = Character("Ogre", hp_max=150, damage=15)
conan = Warrior("Conan", hp_max=120, weapon=sword)
knight = Warrior("Knight", hp_max=100, armor=5, weapon=warhammer)
tank = Warrior("Tank", hp_max=180, damage=8, armor=5)
merlin = Mage("Merlin", weapon=staff)
zara = Mage("Zara", hp_max=70, damage=14, mana=50, mana_max=50)
gandalf = Mage("Gandalf", hp_max=90, mana=20, mana_max=40, weapon=axe)

# Create fight function that simulates a battle and prints the winner
def fight(hero: Character, enemy: Character) -> None:
    Battle(hero, enemy).simulate_battle()
    winner = hero if hero.is_alive else enemy
    print(f"{hero.name} vs {enemy.name}: {winner.name} wins")
    print(f"{hero.name}: {hero.hp} HP left, {enemy.name}: {enemy.hp} HP left\n")

# Simulate fights
fight(alice, bob)
fight(ogre, goblin)
fight(conan, ogre)
fight(knight, tank)
fight(tank, bob)
fight(merlin, alice)
fight(zara, gandalf)
fight(merlin, conan)
fight(zara, knight)
```

```
Alice vs Bob: Alice wins
Alice: 16 HP left, Bob: 0 HP left

Ogre vs Goblin: Ogre wins
Ogre: 138 HP left, Goblin: 0 HP left

Conan vs Ogre: Ogre wins
Conan: 0 HP left, Ogre: 88 HP left

Knight vs Tank: Knight wins
Knight: 40 HP left, Tank: 0 HP left

Tank vs Bob: Bob wins
Tank: 0 HP left, Bob: 0 HP left

Merlin vs Alice: Merlin wins
Merlin: 100 HP left, Alice: 0 HP left

Zara vs Gandalf: Zara wins
Zara: 27 HP left, Gandalf: 0 HP left

Merlin vs Conan: Merlin wins
Merlin: 100 HP left, Conan: 0 HP left

Zara vs Knight: Zara wins
Zara: 16 HP left, Knight: 0 HP left
```

Gen AI-use

Thea:

- Search for bugs in final code
- Used gen AI to generate the code for the battle simulations, and then tweaked it a little bit

Kiki:

- how to use Weapon (class) as an argument in Character *init*
- debugging help with attach method in Character class.
- how to import class from different file

Panopto-video

https://nmbu.cloud.panopto.eu/Panopto/Pages/Viewer.aspx?id=557ec6f8-79de-4186-936d-b4f900c759f3